

North East University Bangladesh

Department of Computer Science and Engineering

Course Title: Deep Learning

Course Code: CSE-06134023

Submitted To:

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1. Title:

Breast Cancer Detection using Artificial Neural Network (ANN).

2. Dataset Used:

This project uses the Breast Cancer Dataset from Scikit-learn, which contains medical diagnostic data for tumor classification.

Key Features include:

Radius

Texture

Perimeter

Area

Smoothness

Compactness

Concavity

Symmetry

Target Variable:

A binary classification variable:

1 (Malignant): Cancerous tumor

0 (Benign): Non-cancerous tumor

3. Methodology:

• Data Preparation:

Required libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn were imported. The dataset was loaded and explored.

• Data Preprocessing:

Checked for missing values and duplicates

Encoded categorical labels into numeric format

Separated features and target variables

• Feature Scaling:

StandardScaler was applied to normalize feature values for better ANN performance.

• Data Splitting:

The dataset was divided into:

80% Training Data

20% Testing Data

• Model Development:

A feedforward Artificial Neural Network (ANN) was built using Keras.

• Training:

The model was trained for multiple epochs using training data.

• Evaluation & Visualization:

Model performance was evaluated using:

Accuracy

Confusion Matrix

Loss and Accuracy Graphs

4. Model Description:

A Feedforward Artificial Neural Network (ANN) was implemented for binary classification.

Architecture:

Input Layer: Accepts all dataset features

Hidden Layers:

Layer 1: 30 neurons (ReLU)

Layer 2: 10 neurons (ReLU)

Output Layer:

1 neuron (Sigmoid activation)

Configuration:

Loss Function: Binary Crossentropy

Optimizer: Adam

Metric: Accuracy

5. Results and Observations:

Performance Metrics:

Accuracy: ~95% (approx, based on ANN performance)

Observations:

The model achieved high accuracy in detecting breast cancer.

ANN successfully classified malignant and benign tumors.

Feature scaling improved model performance significantly.

The model shows strong generalization ability.

Minimal overfitting observed between training and testing results.

6. Conclusion:

This project demonstrates the effective use of an Artificial Neural Network (ANN) for breast cancer detection.

Key Conclusions:

ANN is highly effective for medical classification problems.

Proper preprocessing improves model accuracy.

The model can assist in early cancer detection.

Future Improvements:

Hyperparameter tuning

Use of deeper neural networks

Integration with real-time healthcare systems